Malaysia
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## Summary

Computer Science graduate majoring in Big Data and Artificial Intelligence with hands-on experience in Python, JavaScript, TypeScript, React, FastAPI, REST APIs, and AI integration. Developed an AI-assisted application involving IoT sensor data processing, anomaly detection, API integration, automated reporting, testing, and troubleshooting. Interested in software development, AI-powered applications, IoT, and emerging technologies.

## Experience

**Data Entry Intern**
January 2024 to June 2024
**Noetic Synergy - Puchong, Selangor, Malaysia**

- Maintained and validated enterprise business data in the IFCA system using SQL-supported workflows.
- Identified and corrected missing, duplicated, and inaccurate records to improve data quality and reporting reliability.
- Worked with business data covering contracts, payments, bookings, sales transactions, and reporting.
- Supported senior staff during client-site sessions and helped users understand system workflows and day-to-day usage.
- Gained exposure to how enterprise systems support business operations and reporting.
- Assisted with troubleshooting data and system-related issues and maintained accurate records for reporting.

## Skills

- **Programming & Data**
  Python | SQL | PySpark | Pandas | NumPy | JavaScript | TypeScript
- **Data Engineering & Big Data**
  Apache Spark | Data Processing | Data Cleaning | Data Validation | Data Transformation | Relational Databases
- **AI & Machine Learning**
  OpenAI API | LLM Integration | Prompt Engineering | Machine Learning | Anomaly Detection | Model Evaluation
- **Generative AI Concepts** RAG | Embeddings | AI Agents
- **Backend & Integration**
  FastAPI | REST APIs | API Integration
- **Frontend** React
- **Databases**
  SQLite | Microsoft SQL Server
- **Development Tools**
  Git | GitHub | Debugging | Testing | Documentation

## Education

**Bachelor of Science : Bachelor Of Computer Science, 2026**
**University Of Wollongong - Wollongong, Australia**

- Used Apache Spark and PySpark in university coursework to work with large datasets and perform data processing and analysis.

**BinaryMed - AI-Powered Medical Sample Monitoring System | Final Year Project**
*React, TypeScript, Python, FastAPI, OpenAI API, IoT Data*

- Developed an IoT-based medical sample monitoring system with compliance checking, anomaly detection, and AI-assisted report generation.

- Worked on frontend development, REST API integration, and LLM-based report generation using React, TypeScript, Python, FastAPI, and OpenAI API.
- Collaborated with an industry sponsor to translate business requirements into system features and improve report accuracy and consistency.

**Bachelor of Science : Bachelor Of Computer Science, 2024**
**University Of Wollongong - Shah Alam, Malaysia**

## Languages

**English:** C1
Advanced

**Chinese (Mandarin):** C1
Advanced

**Malay:** B2
Upper Intermediate

## Projects

**FINAL YEAR PROJECT**
**BinaryMed – AI-Powered Medical Sample Monitoring System**
**https://github.com/BinaryMedAI**
Python | FastAPI | React | TypeScript | OpenAI API | REST APIs | IoT Data

- Developed frontend components for an AI-assisted medical sample monitoring application that processed IoT temperature data to identify anomalies and check compliance.
- Integrated React frontend components with Python/FastAPI backend services through REST APIs.
- Integrated the OpenAI API to generate reports from processed sensor data.
- Tested and investigated data-flow issues, API response inconsistencies and incorrect generated reports to improve system reliability.
- Collaborated with an industry sponsor to understand requirements and translate them into functional application features.

**Personal Project**
**Personal Portfolio Website**
React | TypeScript | GitHub Pages | Responsive Web Design | AI Chatbot
https://henrylum27.github.io/

- Built a personal portfolio website to present my technical skills, work experience, education, and projects in one place.
- Added dedicated project sections for BinaryMed and Data Analyst Agent, including project descriptions, technologies, demos, and GitHub links.
- Included a recruiter-focused AI chatbot that answers questions about my skills, experience, availability, and projects.
- Designed the site to be responsive and easy to use across desktop and mobile devices.
- Deployed the website using GitHub Pages.